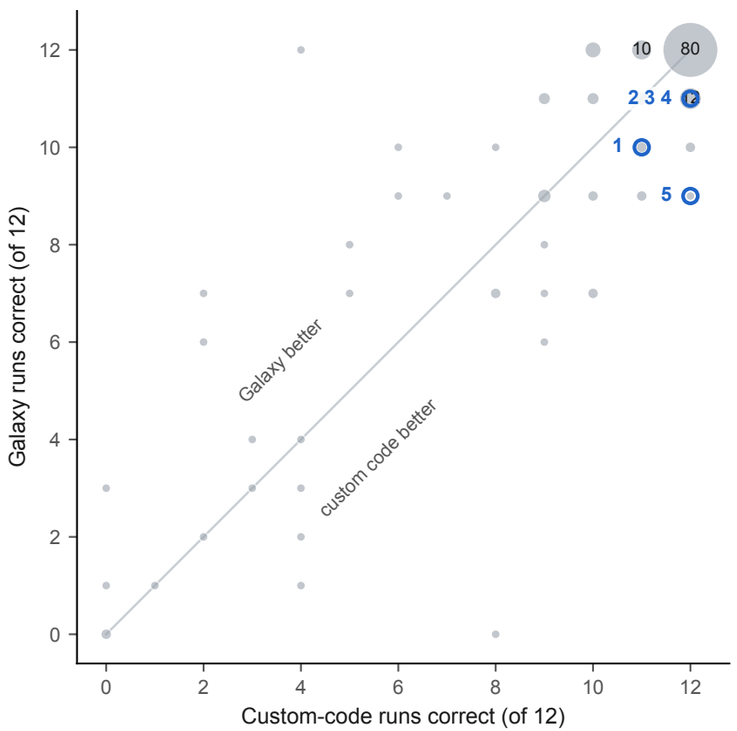
When Galaxy was the problem: custom code solved the task, and Galaxy execution or a Galaxy tool caused the misses

The tasks where the trace audit attributes the Galaxy misses to Galaxy itself. Custom code solved each in 11 or 12 of 12 runs; Galaxy lost one to three runs per task.

a Every task: runs correct in each condition



160 tasks: Galaxy solved fewer runs than custom code on 32, the same number on 94 and more on 34. Bubble area: number of tasks; circled and numbered: the tasks in b.

b The 24 runs of each task: green correct, red wrong; numbers are failed Galaxy jobs in the run

Galaxy execution failed, and the agent switched to a route that gave a different answer

1 bix-31-q2

BixBench-Verified-50. Shrunken log2 fold change of FAM138A with PyDESeq2

Custom code 11/12 **Galaxy 10/12**

	Galaxy			Custom code		
GPT-5.5			1			
Sol			1			
Luna	2	1	2			
DeepSeek		✗	3			✗

DeepSeek r2, r3: the PyDESeq2 UDT jobs never ran (r2) or wrote empty outputs (r3). Both fell back to Galaxy's R DESeq2, giving −0.081 and −0.083 (accepted −0.07 to −0.05). The 10 correct Galaxy runs got a PyDESeq2 UDT to run, some after one to four failed versions.

2 spatial-sim-q2

CompBioBench. Cell types biased toward tumour-core spots

Custom code 12/12 **Galaxy 11/12**

	Galaxy			Custom code		
GPT-5.5	1		6			
Sol		16				
Luna		14				
DeepSeek			17			

Sol r2: every UDT failed to start ("custom project image is not schedulable"). It rebuilt the analysis from about 75 stock-tool jobs and used an adjacency test, which found one of the two cell types. Luna r2 also had 14 failed jobs and recovered.

3 gtf-5-utr-median-len-q1

CompBioBench. Median 5′ UTR, CDS and 3′ UTR lengths of MANE transcripts

Custom code 12/12 **Galaxy 11/12**

	Galaxy			Custom code		
GPT-5.5	2	6				
Sol	1		3			
Luna	3	4	13			
DeepSeek		1	9			

DeepSeek r3: its UDT probes failed, so it rebuilt the task with stock interval tools. The join ignored transcript identity and added 6 bp to two of the three medians.

A Galaxy tool returned a misleading value, and the agent did not check it

4 bix-28-q3

BixBench-Verified-50. Median PhyKIT long-branch score of a gene tree

Custom code 12/12 **Galaxy 11/12**

	Galaxy			Custom code		
GPT-5.5	1					
Sol	✗					
Luna						
DeepSeek						

Sol r1: the PhyKIT wrapper's single non-verbose value is the variance (146.30), not the median (−30.46), and the run submitted it. Two DeepSeek runs saw the same value and rejected it.

5 bix-52-q7

BixBench-Verified-50. Rows removed by a methylation filter

Custom code 12/12 **Galaxy 9/12**

	Galaxy			Custom code		
GPT-5.5						
Sol			✗			
Luna		✗	1			
DeepSeek						

Luna r2, Sol r3: Galaxy's filtered table keeps its header line, and the line count included it (19,160 instead of 19,159). Luna r3 reported the rows kept instead of removed, an agent error.

Sources: task-level trace audit (docs/reports/individual_error_analysis.md); grades as on the results site; failed Galaxy jobs from the run execution-error table (no record for DeepSeek r2 on bix-31-q2, whose jobs never left the queue).